

Report on the Carnegie Research Workshop: *'Continental Climate Change: Simple Models to Understand the Future'*

6-8th June 2022, University of St Andrews

Organisers

- Dr Michael Byrne, University of St Andrews & University of Oxford
- Professor Gabriele Hegerl FRS FRSE, University of Edinburgh
- Professor Jacob Scheff, University of North Carolina at Charlotte



Overview

Understanding how climate change manifests over continents and impacts land-based societies and ecosystems is a critical question in a rapidly warming world. Continental climate is influenced by a complex array of processes, ranging from soil moisture and vegetation to large-scale atmospheric dynamics and interactions with neighbouring oceanic regions. Perhaps because of this complexity, continental climate research has traditionally been undertaken by scientists from different disciplines that rarely interact with each other: climate scientists, hydrologists, meteorologists, plant ecologists, geographers and physicists. The Carnegie Research Workshop on 'Continental Climate Change: Simple Models to Understand the Future' was designed to bring together this wide range of disciplines to advance our understanding of continental climate, with a particular focus on using process understanding based on simple theories and idealised simulations to gain new insights into longstanding problems in the field and explore new avenues & interdisciplinary collaborations for future research.

Structure

30 scientists from 10 countries – largely in the early to intermediate stages of their careers – gathered at the University of St Andrews in early June to attend the 3-day workshop (in addition to 5 virtual attendees). For the vast majority of attendees, this was the first in-person workshop or conference since the Covid-19 pandemic. As such, the workshop provided a timely opportunity for in-depth interactions and discussions among a group of scientists that had not before been assembled. Each day of the workshop followed a similar format, with 30-minute talks (23-minute presentation followed by 7 minutes of feedback/questions) interspersed with dedicated discussion sessions on specific topics. The discussion sessions were largely led by early-career scientists and focused on defining the key questions in continental climate over the coming 5-10 years. The formal workshop agenda was supplemented by networking events – including a group walk along West Sands, a reception at the Wardlaw Museum and a workshop dinner – designed to foster new interactions, connections and collaborations across the group. To engage the broader St Andrew community in the workshop, there was also a vibrant public lecture by Professor Steven Sherwood on ‘Climate Change & Climate Enigmas’ attended by over 100 people including University employees & students, St Andrews residents and students from the local high school.

Session 1: Fundamental Role of Land in the Climate System

The workshop programme was arranged into themes focused on different aspects of continental climate. Monday kicked off with a series of talks on the ‘Fundamental Role of Land in the Climate System’ (Session 1). The presentations addressed deceptively simple yet unsolved questions, including: how do the size and location of continents influence global water vapour and, consequently, planetary temperature (Marysa Laguë); what physical processes control seasonal and diurnal land temperature variations at low (Ori Adam) and high latitudes (Matthew Henry); and how do the laws of thermodynamics constrain the response of continents to climate change (Axel Kleidon). To address these fundamental questions, the speakers presented results from climate simulations with idealised continental configurations along with pencil-and-paper theoretical calculations and observational analyses.

Session 2: Land, Monsoons & Precipitation

The physical controls on tropical precipitation, including monsoons, and their response to climate change are longstanding questions in climate dynamics, and Session 2 on ‘Land, Monsoons & Precipitation’ tackled this problem from a variety of perspectives. Using satellite observations and a novel analytical theory, Cathy Hohenegger and Luca Schmidt investigated the partitioning of tropical precipitation between land and ocean regions. Tropical continents are wetter than would be expected through geographical considerations alone, implying that land-surface properties enhance continental rainfall through their effects on the width and migration of the tropical rainbelt. The influence of a tropical continent on precipitation over neighbouring ocean was investigated using a suite of simulations with an idealised continental configuration (Michela Biasutti). Through advection of dry air, the continent cools and ventilates the oceanic climate to the west, representing a strong non-local influence on the tropical oceanic rainbelt. The behaviour of monsoon precipitation across a wide range of climates was explored using ‘aquaplanet’ simulations (Simona Bordoni). Strikingly, the timing of monsoon onset was found to be progressively delayed as global temperature increases,

with important implications for monsoonal regions in a warming climate. A promising new theory for tropical land precipitation in a changing climate was also introduced (Hugo Lambert), with the theory critically predicting both the ‘thermodynamic’ component of the response – associated with changes in humidity – and the ‘dynamic’ component associated with changes in tropical circulation. Monday’s programme of talks ended with a thought-provoking presentation by Marian Scott on leveraging increasing volumes of data to tackle climate and environmental challenges. On Tuesday morning, Session 2 concluded with presentations investigating how land evaporation (Ruth Geen), vegetation and soil moisture (Marianne Pietschnig) couple to tropical rainfall. Land is complex and difficult to model; these studies elegantly highlighted how simple representations of land-surface processes in idealised climate models can give important new insights into tropical precipitation.

Session 3: The Land Surface & Hydrology

The theme of understanding complex interactions between climate and the land surface was central to Session 3 on ‘The Land Surface & Hydrology’. The scene was set by Steven Sherwood with a review of the various metrics of “aridity”. The proliferation of aridity metrics – including the Palmer Drought Severity Index, relative humidity, runoff, soil moisture and others, which respond to climate change in very different ways – contributes to confusion over whether we expect continents to become “wetter” or “drier” in the future. Sherwood argued that progress in understanding the evolving continental water cycle would be boosted by focusing on *physical measures* of aridity that are connected to the wider climate through fundamental laws (e.g. relative humidity) rather than on metrics defined by the *impacts* of aridity (e.g. crop losses). The question of how climate-driven precipitation changes are partitioned between runoff, evapotranspiration, and soil moisture was then explored in detail using novel climate model analyses (Justin Mankin) and land-surface model simulations (Jack Scheff). The opposing soil moisture trends in the western United States over recent decades in observations versus climate models was investigated by Lucas Vargas Zeppetello, with stomatal closure by plants identified as a possible explanation for the discrepancy. Next, Alexis Berg posed an intriguing question: Why does the continental water cycle respond asymmetrically to a warming versus a cooling climate? Using simulations and theory based on the Budyko curve, Berg argued that changes in vegetation are key to explaining the asymmetric response. Focus then shifted to droughts, and in particular to how these extreme dry events affect terrestrial uptake of carbon from the atmosphere (Vincent Humphrey) and evapotranspiration from soil and vegetation (Alexandra Konings). Evapotranspiration was also the focus of Kaighin McColl’s talk, in which he introduced an elegant new theory to accurately estimate this critical water flux using only weather station data. Currently, evapotranspiration measurements are limited to a small number of FLUXNET sites, so the theory offers a major opportunity to radically expand our measurement capabilities and, consequently, our understanding of the continental water cycle.

Session 4: Extreme Events over Land

Day #3 of the workshop began with a series of talks on extreme weather events over land. The potential use of observed trends and variability in extreme events to constrain future projections was the focus of the first talk (Gabriele Hegerl). New theories to understand the physical mechanisms controlling extreme temperatures in the tropics (Michael Byrne) and extratropics (Yi Zhang) in a warming climate were introduced, along with a discussion of the connections and feedbacks between droughts and heatwaves (Sonia Seneviratne). An

observational analysis revealed, surprisingly, that heatwaves are becoming drier in much of the global subtropics, with important implications for future warming trends (Karen McKinnon). Focus then shifted to extreme rainfall events. Aiguo Dai explored how the energy available to fuel thunderstorms changes as climate warms, and Angeline Pendergrass discussed a simple “two mode” framework to interpret how the probability distribution of rainfall responds to a warming climate. Finally, the behaviour of extreme rainfall over the highly-populated Indian subcontinent during the monsoon season was found to be strongly modulated by the El Niño-Southern Oscillation, with potentially important implications for future changes (Spencer Hill).

Sessions 5 and 6: Computational Techniques & Modelling / Historical Climate Change

The final afternoon of the workshop featured a series of presentations exploring how a range of statistical and modelling techniques can give insights into continental climate. Manoj Joshi used multi-linear regression, a random forest algorithm and satellite measurements to predict the urban heat island effect and its dependence on surface conditions. Novel climate simulations and analysis methods – in particular perturbed parameter ensembles combined with causal inference methods (David Sexton), particle filter experiments (Andrew Schurer) and a linear inverse model (Flavio Lehner) – were applied to quantify and understand the influences of model uncertainties, forcings and natural variability on historical climate trends.

Summary and Next Steps

By design, a diverse range of topics and scientific techniques were presented over the three days of the workshop. A unifying theme – emerging from across the talks and discussion sessions – was that, in order to advance understanding of continental climate, the community needs a renewed focus on developing simple theories and conceptual frameworks to complement state-of-the-art, high-resolution climate models and statistical/machine learning techniques. Priority areas where simple theories have potential to provide new insights, highlighted during the workshop, include the regional water cycle (specifically the role of the large-scale circulation), soil moisture and atmosphere-vegetation interactions. In bringing together existing and emerging leaders in the field, the workshop has created a new community focused on accelerating research on questions in continental climate that go back decades, using a diverse and complementary range of research approaches. A perspective article, setting out the vision of this new community for the field over the next decade, is currently being planned and will be the key output of this Carnegie Research Workshop. Finally, the workshop was very well received by the attendees, with a professor at a leading US university commenting afterwards that “[i]t was probably the best workshop or conference I've ever attended.” Given the success of the meeting, it is anticipated that this Carnegie Research Workshop will inspire a follow-up event on continental climate in the coming years.

List of attendees (early-career scientists underlined)

Name	Institutional affiliation	Job title
Adam, Ori	Hebrew U.	Senior lecturer
Berg, Alexis	U. Montreal	Assistant professor
Biasutti, Michela	Lamont-Doherty Earth Observatory	Research professor
Bordoni, Simona	U. Trento	Professor
Byrne, Michael	U. St Andrews	Reader
<u>Chaudhri, Alex</u>	U. St Andrews	PhD student
Dai, Aiguo	U. Albany	Professor
<u>Duffield, Josh</u>	U. St Andrews	Incoming PhD student
Geen, Ruth	U. Birmingham	Lecturer
Hegerl, Gabi	U. Edinburgh	Professor
<u>Henry, Matthew</u>	U. Exeter	Postdoctoral researcher
<u>Hill, Spencer</u>	Columbia U.	Research scientist
Hohenegger, Cathy	Max Planck Institute for Meteorology	Group leader
<u>Humphrey, Vincent</u>	ETH Zürich	Postdoctoral researcher
Joshi, Manoj	U. East Anglia	Professor
Jump, Alistair	U. Stirling	Professor
Kleidon, Axel	Max Planck Institute for Biogeochemistry	Group leader
Konings, Alexandra	Stanford U.	Assistant professor
<u>Laguë, Marysa</u>	U. Saskatchewan	Postdoctoral researcher
Lambert, Hugo	U. Exeter	Associate professor
Lehner, Flavio	Cornell U.	Assistant professor
Mankin, Justin	Dartmouth College	Assistant professor
McColl, Kaighin	Harvard U.	Assistant professor
McKinnon, Karen	UCLA	Assistant professor
Pendergrass, Angie	Cornell U.	Assistant professor
<u>Pietschnig, Marianne</u>	U. Oslo	Postdoctoral researcher
Scheff, Jack	UNC Charlotte	Assistant professor
<u>Schmidt, Luca</u>	Max Planck Institute for Meteorology	PhD student
<u>Schurer, Andrew</u>	U. Edinburgh	Chancellor's fellow
Scott, Marian	U. Glasgow	Professor
Seneviratne, Sonia	ETH Zürich	Professor
Sexton, David	UK Met Office	Head of Scenarios Dev.
Sherwood, Steven	U. New South Wales	Professor
<u>Van De Koot, Emily</u>	U. Oxford	PhD student
<u>Vargas Zeppetello, Lucas</u>	U. Washington	Postdoctoral researcher
<u>Zhang, Yi</u>	UC Berkeley	Postdoctoral researcher